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15 September 1969
SUPERSEDING
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MILITARY SPECIFICATION

ANTENNA: AS-2236, 30 TO 76 MEGAHERTZ

1. SCOPE

1.1 This specification covers requirements for one type antenna, man or vehicular transportable.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment.
MIL-E-5272	-	Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-15024	-	Plate, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore, General Specification.
MIL-E-17555	-	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-E-21981	-	Electronic Type Designations, Identification Plates and Markings, Requirements for.

FSC 5985

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(Copies of specifications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 First article. The first article is defined as an antenna which will be identical to the product which the supplier will subsequently produce in fulfillment of the contract. Unless otherwise specified (see 6.2), as soon as practicable after award of the contract or order and prior to submission of any completed units for final acceptance, the contractor shall furnish four first articles for testing under the direction and supervision of the cognizant Government inspector to determine conformance to this specification. Examination and tests shall be as specified herein. Approval of the first article by the activity concerned shall not relieve the contractor of the responsibility to supply equipment conforming to this specification. Any changes or deviations of the production units from the first articles shall be subject to the approval of the contracting officer. The first articles may be included in final shipment as part of the total quantity to be furnished under contract or order, provided they are completely refurbished.

*3.2 Parts and materials. In the selection of parts and materials, fulfillment of major objectives shall be the prime consideration. In all cases, general material requirements as to flammability, fungus-resistance and nontoxicity as required by MIL-E-16400 shall be met.

*3.2.1 Mechanical parts. Where mechanical parts whose function is to rotate, cam or oscillate against others are employed, and the best choice of materials includes other than those listed in MIL-E-16400, use of the other materials may take precedence. The contractor shall submit evidence that improved maximum life will be achieved by use of lubricating agents, protective paints, sprays or other methods.

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*3.2.2 Nonstandard parts. When it has been determined that requirements cannot be met by use of parts and materials as specified in MIL-E-16400, immediate action shall be taken by the contractor to inform the contracting officer by submission of detailed data, as required by MIL-E-16400, requesting approval.

*3.2.3 Fungus treatment. The equipment shall be fungus proofed by selection of parts and materials which are nonnutrient for fungus, or the parts and materials used shall be treated in accordance with MIL-T-152 prior to use in the equipment.

*3.3 Design and construction. The antenna, AS-2236, shall consist of a log periodic directional antenna, a vertically erected mast, a tripod assembly and the necessary ancillary equipment to provide a complete antenna. It shall be capable of providing directional antenna performance. The antenna shall be capable of handling all radio frequency power levels up to at least 65 watts, continuous wave, over the 30 to 76 megahertz (MHz) spectrum. The antenna shall consist of the following:

Item No.	Item	Qty.
1	Tripod assembly	1
2	Mast assembly	1
3	Antenna assembly	1
4	Guy cable assembly	1
5	Guy anchor, set	1
6	Tripod anchor, set	1
7	Carrying cases, set	1
8	Radio frequency (rf) cable assembly	1
9	Earth anchor setting implement	1
10	Polarizer	1
11	Tripod Brace	1

*3.3.1 Tripod assembly. The tripod assembly shall be a lightweight unit consisting of a mast launching assembly and tripod base. The launching assembly shall provide guides for the mast section during erection of the antenna. A vertical restraint device shall be provided to support the antenna mast assembly in the extended position. The

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launching section shall be equipped with a clamping mechanism for maintaining azimuth orientation of the antenna, while providing for intermediate vertical restraint of the antenna. The launching section shall be capable of supporting the antenna at any incremental height up to its fully extended position. The tripod assembly shall be sufficiently rigid to support the antenna at its fully extended height of 35 feet in horizontal wind conditions of 20 knots, without guying. No damage shall be sustained by any component of the antenna during wind conditions up to 60 knots, horizontal, at its fully extended height, when the guy cable assembly is employed.

*3.3.2 Tripod legs. The tripod shall consist of three folding, telescoping tripod legs. A minimum of 12 inches of telescopic adjustment shall be provided in each leg to permit leveling the base on uneven terrain. Locking of each leg at its desired height shall be accomplished with the use of a hand actuated device. No tools shall be required. The legs shall be provided with drilled, swiveled feet to allow staking the tripod into the earth. The foot pads of the tripod shall measure between 8 and 9 inches square. The tripod shall be capable of erection, adjustment and disassembly for transport without the use of tools.

*3.3.3 Tripod brace. A means shall be provided for connecting the lower portion of the tripod legs to each other to form a triangular truss. The device is intended to prevent the tripod footpads from sliding with respect to each other when soil conditions reduce the effectiveness of footpad stakes. The device is to consist of flexible steel cables with a tension adjusting device.

*3.3.4 Mast assembly. The mast assembly shall consist of eight readily detachable, mechanically interchangeable sections of such dimensions as to adequately support the antenna at its fully extended height of 35 feet.

*3.3.4.1 Lower mast sections. The six lower mast sections shall be provided with keyed couplings so as to provide positive azimuth orientation of the antenna.

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*3.3.4.2 Upper mast sections. The two upper mast sections shall be made of a dielectric material so as to prevent distortion of the antenna pattern when the antenna is placed in either the horizontal or vertical plane. They shall be provided with keyed couplings so as to maintain positive azimuth orientation of the antenna. The dielectric mast sections shall provide accommodation for the attachment of the guy cable assembly.

*3.3.5 Guy cable assembly. The guy cable assembly shall consist of three dielectric rope guy lines, tensioning devices and mast mounting hardware sufficient to maintain the antenna upright in a 60-knot wind when in its fully extended position.

*3.3.6 Earth anchors.

*3.3.6.1 Guy anchor set. Three auger-type earth anchors shall be provided to fasten the guy cables. They shall be at least 18 inches in length and have a helix plate 3 inches in diameter. A hook shall be provided in the top of the shaft for attachment of the guy cables. The auger shall be capable of resisting a 300-pound force when installed with the antenna in loose, sandy soil.

*3,3.6.2 Tripod anchor set. Six spike-shaped stakes shall be provided for holding the tripod footpads in place. The stakes shall fit through the holes in the footpads and have a head or other feature to capture the footpad securely in place. These stakes shall be approved by the procuring activity.

*3.3.7 Carrying case set. Three carrying cases shall be provided for the purpose of transporting all components of the antenna. The carrying cases shall be provided with carrying handles and shoulder straps. An accessory bag, or suitable substitute, which attaches to, or stows within, one of the carrying cases, shall be provided.

*3.3.8 Radio frequency cable assembly. The rf cable assembly shall be 125 feet in length, constructed of RG-8A/U coaxial cable, and terminated with a type BNC male connector on each end. A clamp or similar device shall be installed at the upper cable end of the mast to eliminate strain on the connector caused by the weight of the dangling cable.

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*3.3.9 Polarizer. A device shall be provided which permits the polarization of the antenna to be changed without disassembling the mast or removing any guy lines. The device shall be mounted atop the mast just below the antenna and may be operated by ropes hanging down to the ground. The design shall insure that polarization position cannot be changed accidentally by wind forces.

*3.4 Reliability. Reliability shall receive major emphasis during the design, fabrication and construction of antennas, masts and ancillary equipment. This emphasis shall be reflected in the assembly processes and the determination of operational techniques. The first articles and selected sample equipments shall be subjected to complete assembly, erection and disassembly 100 times immediately prior to the mechanical and electrical tests required by Section 4 of this specification. Mechanical wear shall be such that mechanical or functional performance is not affected. Abrasion of painted surfaces is allowable; grooving or cutting of cammed, joined or sliding metal surfaces, or cracking or breaking of any portion of the antenna shall not be allowable. Electrical performance of the antenna shall not be affected in any way. Earth anchor bending, breaking or deformation is allowable.

*3.5 Performance. The antenna shall be a lightweight, tactical type, high performance antenna covering the frequency range of 30 to 76 MHz. The antenna shall provide the following:

(a) Transmission and reception of horizontally or vertically polarized rf signals without tuning of the antenna, transmission line or matching device.

(b) Capable of being erected by two men in less than 15 minutes.

(c) Simple azimuth orientation.

(d) Capable of operation in winds of up to 33 knots without guying, or 60 knots with guying provided as part of this antenna at its fully erected height of 35 feet.

(e) A minimum of 4.5 decibels (dB) isotropic gain.

(f) A minimum front-to-back ratio of 8 dB.

(g) A maximum voltage standing wave ratio (vswr) of 3:1.

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(h) A maximum total weight of not more than 115 pounds and packaged so that it may be easily transported by three men.

*3.5.1 Electrical performance. The electrical characteristics of the antenna shall be in accordance with this specification. All characteristics stated shall be required when the antenna is erected at its fully extended height of 35 feet above the terrain, in either the horizontal or vertical plane.

*3.5.1.1 Forward gain. The antenna shall exhibit not less than 4.5 dB forward isotropic gain over the frequency range of 30 to 35 MHz and not less than 5.0 dB forward isotropic gain over the frequency range of 35 to 76 MHz.

*3.5.1.2 Characteristic impedance. The antenna shall present a nominal load of 50 ohms, measured at the input connector, over the frequency range of 30 to 76 MHz.

*3.5.1.3 Front-to-back ratio. The antenna shall provide a front-to-back ratio of not less than 7.5 dB over the frequency range of 30 to 76 MHz.

*3.5.1.4 Voltage standing wave ratio. The antenna shall present a vswr, measured at the input connector of the rf cable assembly, of not greater than 3:1 over the frequency range of 30 to 76 MHz.

*3.6 Dimensions. Dimensions of major components shall be as follows:

(a) Tripod assembly, when collapsed and packaged for transport, external dimensions shall not exceed 15 inches by 48 inches maximum.

(b) Mast assembly, when collapsed and packaged for transport, external dimensions shall not exceed 12 inches by 48 inches maximum.

(c) Antenna assembly, when collapsed and packaged for transport, external dimensions shall not exceed 10 inches by 48 inches maximum.

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*3.7 Weight. The maximum weights of the antenna components shall be as follows:

Item No.	Item	Max. Weight, Pounds
1	Tripod assembly	21
2	Mast assembly	27
3	Antenna assembly	15
4	Guy cable assembly	3-1/2
5	Guy anchor, set	4-1/2
6	Tripod anchor, set	3
7	Carrying cases, set	16
8	Radio frequency cable assembly	13-1/2
9	Earth anchor setting implement	3
10	Polarizer	4
11	Tripod Brace	2-1/2

*3.8 Finish. Metal surfaces on the tripod, mast and antenna shall be painted with a semigloss epoxy enamel; color shall match Marine Corps green, No. 23 (see 6.3). Surfaces subject to sliding wear, such as metal mast sections, tripod footpads and earth anchors may, alternatively, be anodized a dark green color in accordance with type II of MIL-A-8625.

*3.9 Identification marking. Subassemblies shall be identified with the following information rubber stamped or silk screened on a fixed surface, if available, or on a permanent tag affixed in such a manner as not to impair operation of the assembly. The marking shall not be easily removable.

(a) Nomenclature
(b) Approved item name
(c) Additional information required; Federal Stock No., Drawing No., manufacturer's name, and serial number.

*3.9.1 Identification plates. Identification plates shall be in accordance with MIL-P-15024 and shall be of aluminum alloy where practicable. Drawings for identification plates shall be submitted for approval 60 days prior to commencement of first article tests. Nomenclature shall be requested in accordance with MIL-E-21981 and shall be submitted 90 days prior to commencement of first article testing.

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*3.10 Workmanship. Antennas covered by this specification shall be manufactured and finished in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspection and tests. Inspection and tests shall be classified as follows:

(a) First article (see 4.2.1)

(b) Production (see 4.2.2)

(c) Sample (see 4.2.3)

*4.2.1 First article. Four antennas shall be used to complete the tests indicated for first article in Table I. First article testing will be witnessed by a representative of the contracting officer.

*4.2.2 Production. All antennas shall be subjected to the tests indicated for production in Table I.

*4.2.3 Sample. Antennas, randomly selected by the cognizant Government inspector in the quantity specified in 4.2.3.1, shall be subjected to the tests indicated for sample in Table I.

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Table I - Tests

Performance	Reqt. Para.	Vibration	Shock	Temperature Shock	Humidity	Altitude	Salt Spray	Rain	Wind Loading	Standard Conditions
Assembly erection and disassembly	3.4									FA S
Forward gain	3.5.1.1			FA S	FA S	FA S	FA S	FA S	FA S	FA S
Characteristic impedance	3.5.1.2				FA S	FA S	FA S	FA S		FA S
Front to back ratio	3.5.1.3				FA S	FA S	FA S	FA S		FA S
Voltage standing wave ratio	3.5.1.4				FA S	FA S	FA S	FA S		FA S P
Erection time	3.5									FA S
Mechanical and visual inspection	various	FA S	FA S	FA S	FA S		FA S		FA S	FA S P
Items to be tested										
Tripod assembly		X	X		X		X	X	X	X
Mast assembly		X	X		X		X	X	X	X
Antenna assembly		X	X	X	X	X	X	X	X	X
Guy cable assembly		X	X				X		X	X
Earth anchors		X	X						X	X
Carrying cases		X	X							X
RF cable assembly		X	X	X	X		X	X	X	X
Earth anchor setting implement		X	X							X

Note: P= Production FA= First article S= Sample

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*4.2.3.1 Sampling plan. Sampling plan quantity shall be selected as follows:

Qty. of equipment produced	Qty. to be tested
1-49	1
50-99	2
100-149	3
150-199	4
200-300	5
301-400	6
	One additional antenna for each 100 produced.

*4.2.3.2 Retest procedures. Failure of the sample shall constitute rejection of the lot. The defective antenna shall be refurbished, returned to the lot, and another random selection made for retest. The retest selection will be double the quantity of the failing, defective antennas.

*4.3 Tests.

*4.3.1 Vibration. Each antenna to be subjected to the vibration test shall be disassembled, stowed in its carrying cases and rigidly mounted to the vibration table in the horizontal plane. It shall be subjected to vibration in the frequency range of 10 to 55 Hertz (Hz) at 5 gravity (g), or 0.06 inches total excursion, whichever is greater. Frequency change shall be at a rate of less than one Hz. Duration of the test shall be one-half hour. There shall be no destructive rigid body resonance exhibited below 55 Hz.

*4.3.2 Shock. Each antenna, with components, to be subjected to the shock test shall be disassembled and stowed in the carrying cases. It shall withstand a 3-hour test on the LAB package tester. For this test, the carrying cases shall be rigidly mounted to the adapter plate. The machine shafts shall be adjusted in speed and operated in phase to produce average peak accelerations of 10 g's with occasional peaks of 20 g's.

*4.3.3 Temperature shock. The equipment shall be tested in accordance with procedure I of MIL-E-5272, except that the high temperature shall be +65°C.

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*4.3.4 Humidity. Antennas shall be tested in accordance with procedure I of MIL-E-5272 except the maximum temperature shall be +65°C. The test will be concluded at the end of three cycles. Any external moisture present may be removed by wiping.

*4.3.5 Altitude. The antenna equipment shall be tested in accordance with procedure VI of MIL-E-5272 to 50,000 feet at -40°C.

*4.3.6 Salt spray. Place equipment in test chamber in disassembled condition and perform this test in accordance with MIL-E-5272. After removal from the test chamber, electrical performance of the antenna shall not be degraded more than 15 percent. When salt accumulation is removed by washing or hosing with fresh water, the antenna electrical performance must meet the standards prescribed in 3.5.1.

*4.3.7 Rain. Place the antenna in its normal operational position. Allow simulated rainfall to fall on the antenna at a minimum rate of 2 inches per hour for 1 hour. Immediately after completion of the simulated rainfall, subject the antenna group to all electrical test requirements of 3.5.1. Performance shall not be degraded below the prescribed standards. All electrical tests must be completed in less than 4 hours after soaking. Resoaking will be allowed.

*4.3.8 Wind. The antenna system shall be subjected to the loading placed on it by a 60-knot wind blowing horizontally, for a period of 1 hour. At the end of this test, the antenna system shall show no evidence of electrical or mechanical degradation. Compliance with this procedure for the antenna mast may be demonstrated by appropriate calculation and simulation.

*4.4 Approval. Approval of the first articles shall be by the contracting officer upon satisfactory completion of all the tests.

*4.5 Acceptance. With exception of the first articles, acceptance of all items shall be by the cognizant Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of equipments shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

*6.1 Intended use. This antenna is to be employed as a directional, lightweight, man-transportable antenna for use with tactical radio equipment operating in the frequency spectrum of 30 to 76 MHz.

*6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Whether first article is required (see 3.1).

(c) Level of preservation, packaging and packing (see 5.1).

6.3 Color. Color chips for Marine Corps green, No. 23 may be obtained from the Commandant of the Marine Corps (Code CSY-10), Headquarters U. S. Marine Corps, Washington, D. C. 20380.

*6.4 First article. First article inspection shall be in accordance with Section 4.

*6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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